

James Zhang

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EDUCATION

Yale University

Sep 2025 – Dec 2025

Visiting Graduate Student

Grades: Honors

Declined a full-time offer at Amazon to explore a PhD in financial economics; returning to industry.

Coursework: Financial Economics I, Econometrics I

University of Maryland, College Park

Aug 2022 – May 2025

B.S. Computer Science & Mathematics, Minor in Computational Finance

GPA: 3.9/4.0

Teaching Assistant: Portfolio Management (BUFN402), Computational Finance Capstone (BUFN403)

Coursework: Design and Analysis of Algorithms, Computer Systems, Computer Networks, Machine Learning, Deep Learning, Probability Theory, Mathematical Statistics, Real Analysis, Abstract Algebra, Honors Multivariable Calculus and Linear Algebra

EXPERIENCE

Independent Algorithmic Trading – Prediction Markets

New Haven, Connecticut

Founder, Research Engineer

Apr 2026 – Present

- Leading the operation and defining the research and trading agenda, splitting development with one other engineer.
- Building and deploying a Rust trading system on AWS that combines systematic market making and signal-based trading across Kalshi crypto and commodity markets, trading 200k+ contracts weekly for \$1,200 profit on \$50/month of cloud infra.
- Using data analysis and historical data to manage a sportsbook that quotes same game parlays on Kalshi, pricing correlated parlays and answering live request-for-quote auctions across NBA, NFL, MLB, and World Cup matches.

Yale University

New Haven, Connecticut

*Research Professional, School of Management
with Bryan Kelly*

Aug 2025 – Present

- Built a large language model pipeline that reads three decades of Dow Jones Newswires and Thomson Reuters articles and returns structured, schema-validated belief annotations, assigning each story to the markets it moves.
- Served Gemma 4 26B-A4B (mixture-of-experts) and Qwen 3.8-27B through vLLM with schema-constrained decoding and batch invariance on Yale's SLURM HPC cluster, running over 30 million inference calls to build a novel text dataset.
- Cut GPU costs and wall time by 5x using prefill cache optimizations and keyword prefiltering before inference.
- Extended the text-as-data finance literature by showing that ridge regression forecasts on news signals earn 9.7% annual alpha ($t=3.0$) on 25 commodity markets, after transaction costs and controls for momentum, value, basis, and basis-momentum.
- Created [Global Macroeconomic Data](#), a Python package that streams 1.5 TB of tick and macro data from LSEG Tick History, Datastream, WRDS, Compustat, OECD, and FRED through Polars into return panels for over 200 global futures markets.

Amazon

Seattle, Washington

Software Development Engineering Intern

Jun 2025 – Aug 2025

- Shipped new video performance metrics such as 7 day click rates and average watch time to the Video Shopping Experiences portal used by over 2 million sellers and vendors, built end to end from SQL to Java to a localized React frontend.
- Evaluated Elasticsearch against AWS Timestream for the portal's latency and cost targets, met with the Timestream team, and presented a migration recommendation to principal engineers and management that the team adopted.
- Received a full-time return offer.

University of Maryland, College Park

College Park, Maryland

*Research Assistant, Department of Finance
with Serhiy Kozak*

Jan 2024 – May 2025

- Wrote a custom Gaussian kernel operation in C++ and CUDA and registered it as a native Jax primitive with JIT, vmap, and autodiff support, so a kernel model of the cross section of stock returns could be estimated on GPU at full scale.
- Implemented orthogonalized alternating least squares and PARAFAC tensor decompositions in Jax on GPUs to factor a large panel of firm characteristics for a three dimensional factor model of returns.

Capital One

McLean, Virginia

Software Engineering Intern

Jun 2024 – Aug 2024

- Shipped a FastAPI Slackbot to production on AWS ECS Fargate that replaced emailed requests to the Data Remediation team with a tracked Slack workflow, so requesters got answers sooner and engineers stopped losing time to email.

PROFILE

Languages: Python, Rust, Java, C++, CUDA, SQL, JavaScript

Frameworks and Libraries: PyTorch, Jax, Transformers, vLLM, Pydantic, Polars, React, FastAPI, NumPy, Pandas

Infrastructure and Tools: AWS (S3, Lambda, EC2, ECS Fargate, Lightsail), SLURM, Linux, Git, Claude Code, Cursor